

ITEA Conference 2024

Pumping the Brakes on Pass-Through: Spatial Competition and Fuel Taxes – Evidence from a Quasi-natural Experiment in Germany

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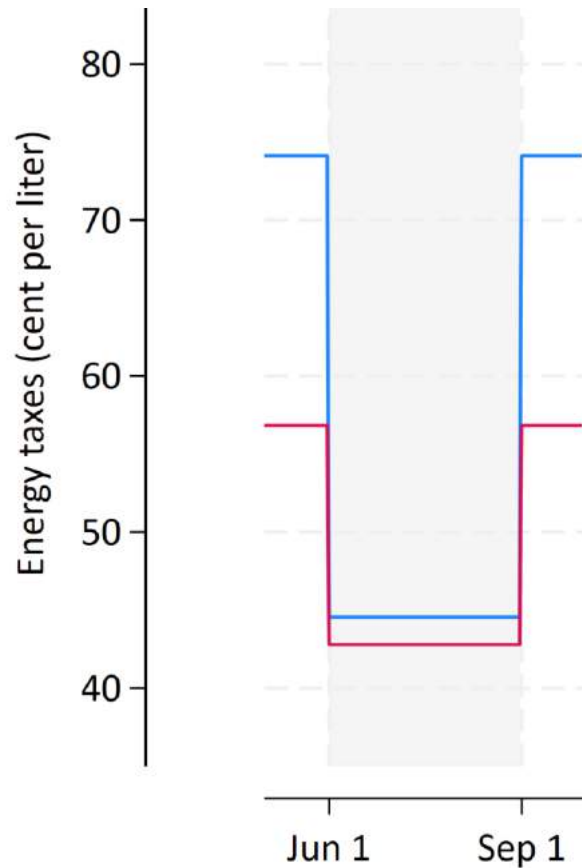
- I. Context
- II. Theoretical Framework
- III. Econometric Approach
- IV. Data
- V. Results
- VI. Further Perspectives

Temporary measures (**June to August**) related to transportation costs to offset the impact of the Russian invasion on private households:

- **„9-euro Ticket“** for public transport
 - Nation-wide unlimited access to buses, subways, and trains for 9 Euro per month

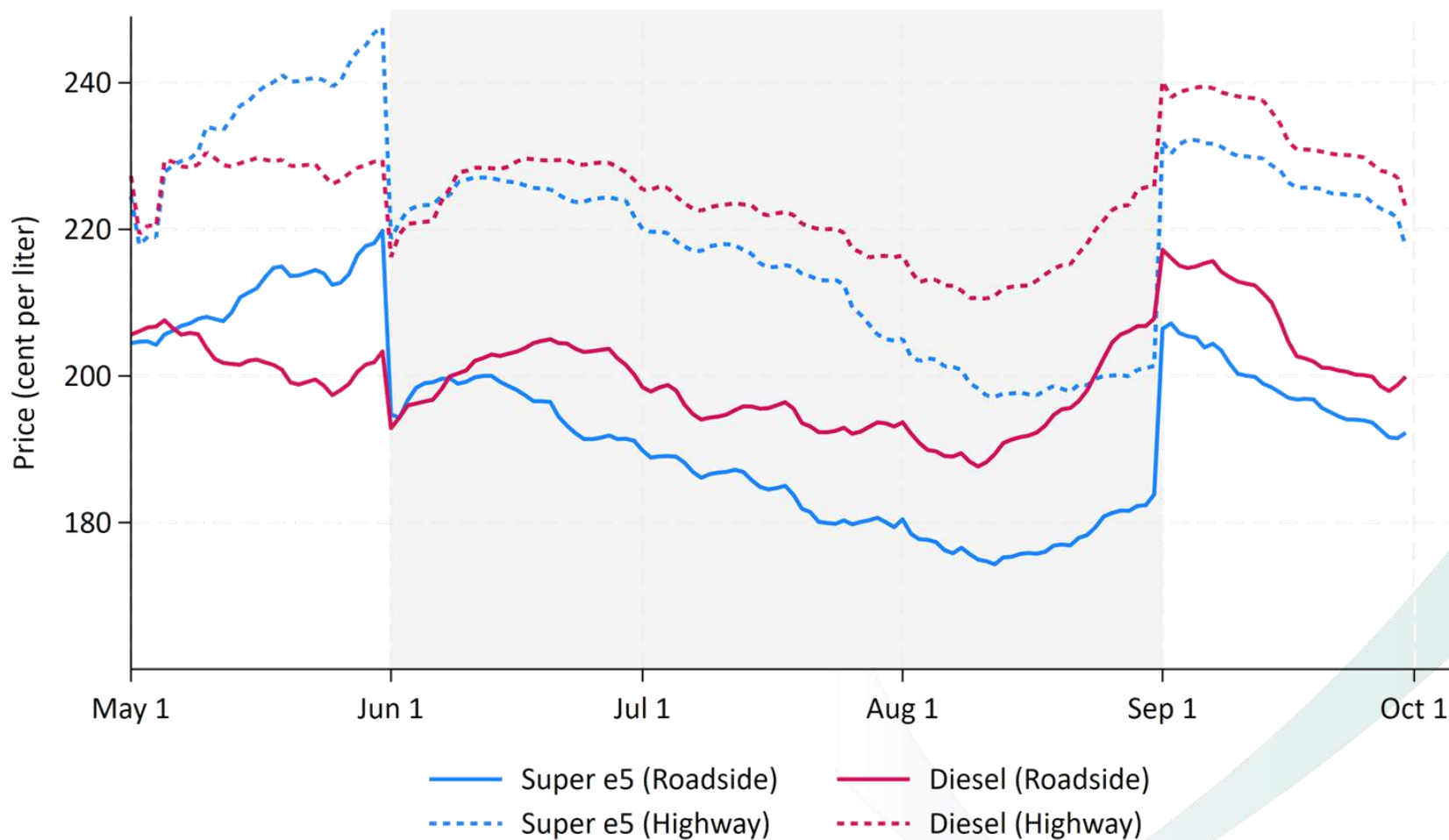
- **Fuel Tax Discount**
 - Reduction of Fuel Taxes to the minimum EU standard

Fuel Taxes and Fees in Germany 2022



	Gasoline (E5 & E10)		Diesel	
Value added tax (VAT)	19 %			
Carbon tax (ct/l)	8.40		9.50	
Fees to the oil reserve (ct/l)	0.27		0.30	
Energy tax (ct/l)	65.45	35.90 (Jun-Aug)	47.04	33.00 (Jun-Aug)
Effective tax discount (including VAT)	35.16		16.71	

Daily Fuel prices in Germany at Roadside and at Highway Stations



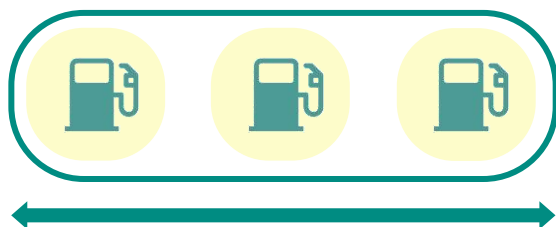
Horizontal Markets

(Jenkin, 1872;
Bulow & Pfleiderer, 1983;
Weyl & Fabinger, 2013)

$$\rho = \frac{dp}{dt} = \frac{1}{1 + \frac{\theta}{\varepsilon_\theta} + \frac{\varepsilon_D - \theta}{\varepsilon_S} + \frac{\theta}{\varepsilon_{ms}}}$$

- Assuming constant marginal costs ($\varepsilon_S \rightarrow +\infty$),
constant conduct parameter ($d\theta/dp = 0$),
and linearity in demand ($\varepsilon_{ms} = 1$):

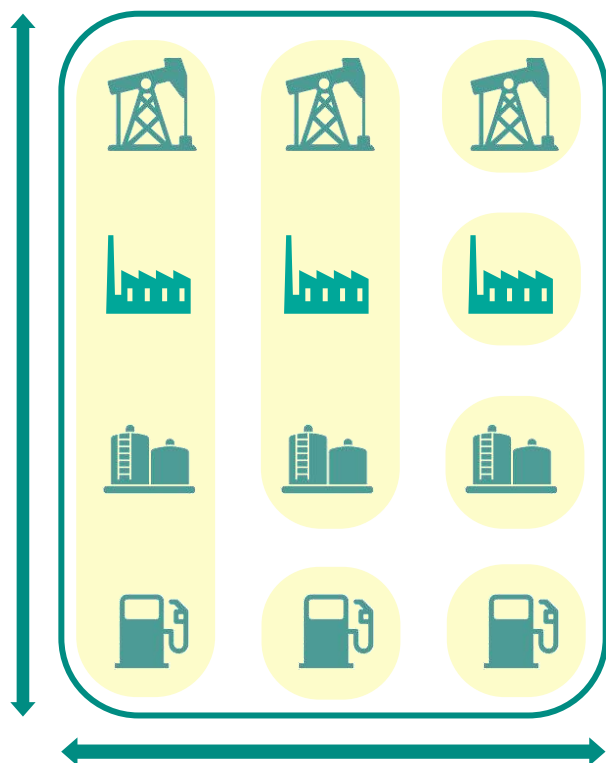
$$\rho = \frac{1}{1+\theta} \quad \rightarrow \quad \rho_I < \rho_C$$



- Pass-Through is predominantly determined by relative supply and demand elasticities
- Under credible assumptions **Imperfect Competition** lowers tax pass-through

Vertical Markets

(Spengler, 1950; Williamson, 1971)



- Different vertical arrangements impact the transmission of cost shocks throughout the supply chain
 - Under linear contracts the **double marginalization** problem arises in imperfect markets
 - Market power at the producer/processing level and retailer level results in markups at each of these stages
- Tax Pass-Through is expected to be higher under **Vertical Integration**
- Double marginalization problem is expected to be mitigated when downstream markets are perfectly competitive

Two-Step Regression Discontinuity in Time (RDiT) Estimation

- „Residualizing“ (Lee & Lemieux, 2010); „Augmented Local Linear“ (Hausman & Rapson, 2018); „Residual Ignorability“ (Sales & Hansen, 2020)
 - 1) Impacts of price influencing factors are estimated using a wide time window, residuals are saved
 - 2) Local linear specification using residuals within narrow bandwidth around treatment
- **First Stage:**

$$p_{ihd} = \alpha + \beta_1 S_{id} + \beta_2 D_{ihd} + hour_h + weekday_d + \mu_i + u_{ihd}$$

for $d \in [70 \text{ days before treatment}, 5 \text{ days after treatment}]$

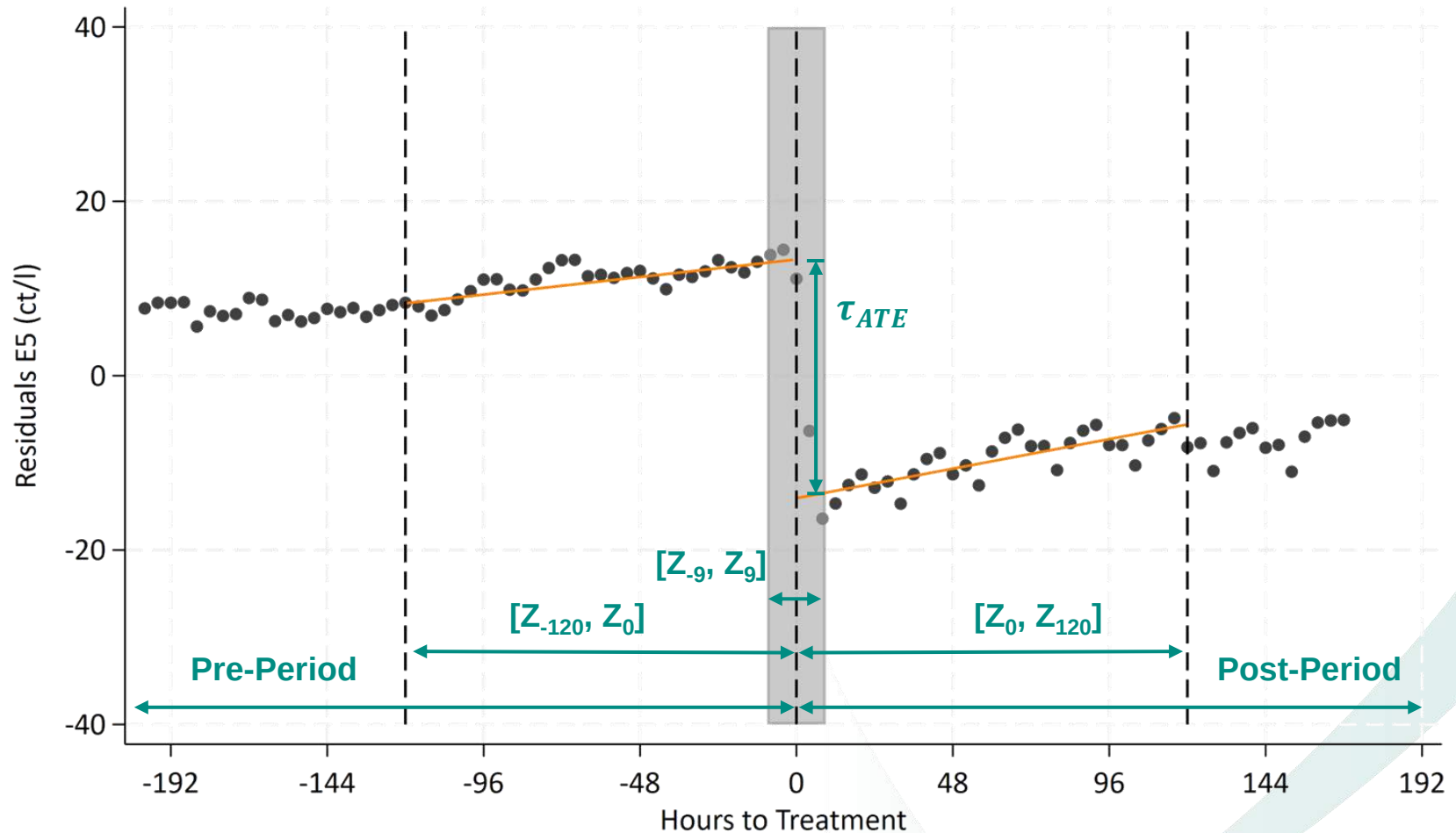
- **Second Stage:**

$$\widehat{u_{ihd}} = \alpha + \beta_1 TR_{hd} + \gamma_1 f(Z_h) + \gamma_2 TR_{hd} \times f(Z_h) + \varepsilon_{ihd}$$

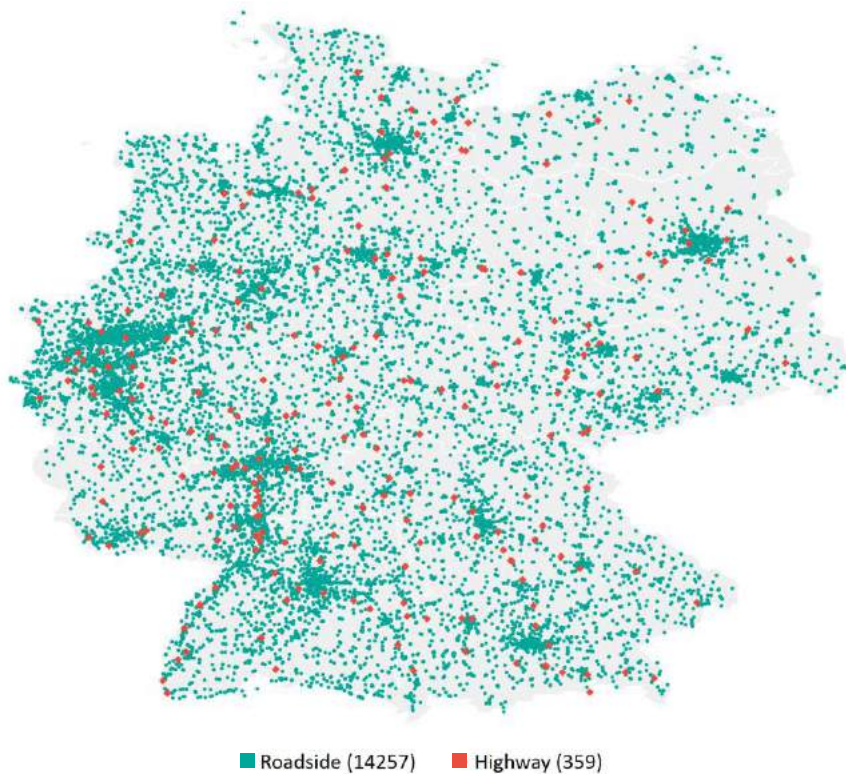
for $Z_h \in [-120, +120]$

$i := \text{station}$
 $h := \text{hour}$
 $d := \text{day}$

Regression Discontinuity in Time (RDiT) Methodology



$$\begin{aligned}\tau_{ATE} &= E[Y_i(1) - Y_i(0) | X_i = c] = \\ &= \lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x]\end{aligned}$$



Pump Prices (Source: Tankerkönig)

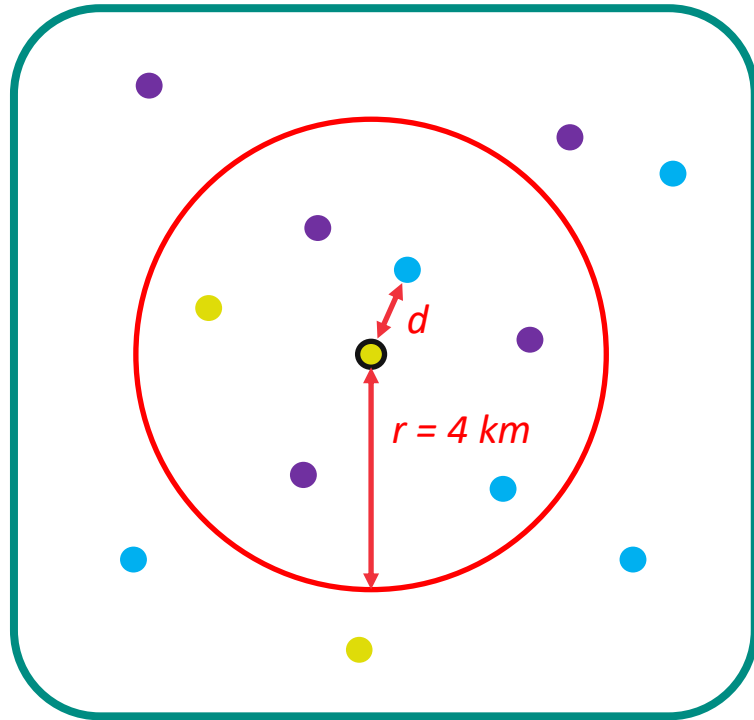
- Price changes with time stamps for gasoline (E5, E10) and diesel

Station Characteristics (Source: Tankerkönig & OSM)

- Station name & brand affiliation
- Openingtimes
- Address
- Geo-coordinates

Further Covariates:

- Supply Shifter
(crude oil prices, exchange rates)
- Demand Shifter
(weather, traffic volumes, holidays)



Indicators for Spatial Competition:

- Distance to the closest competitor d
 - Number of competitors/brands
 - Distribution of market shares
 - Market concentration
- } Based on delineation of local markets (4 km)

Station-level Herfindahl-Hirschman-Index (HHI):

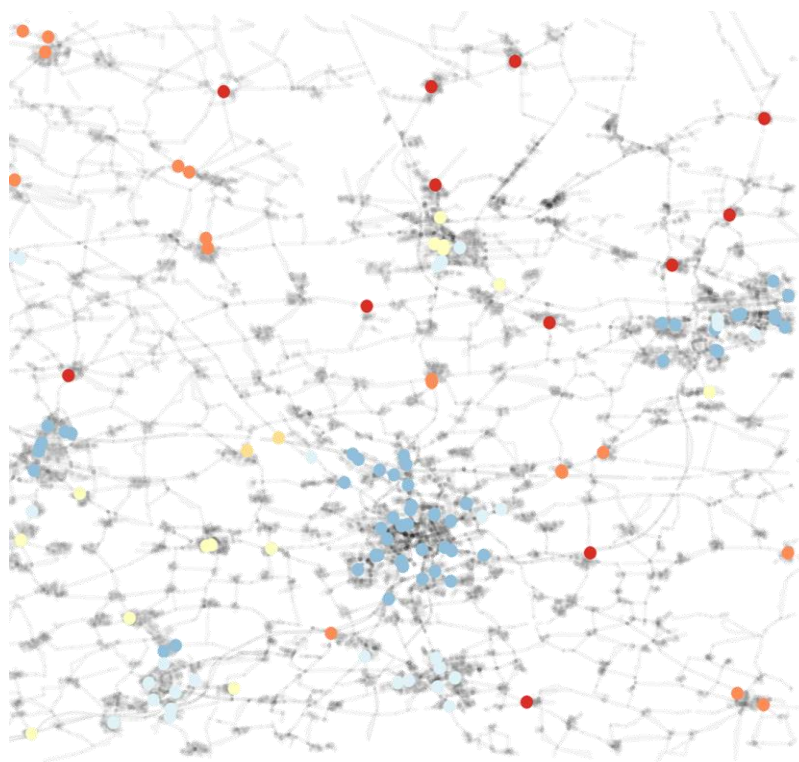
$$HHI_r = \sum_{k=1}^N \left(\frac{x_k}{\sum_{j=1}^N x_j} \right)^2 \quad \forall i$$

i := station

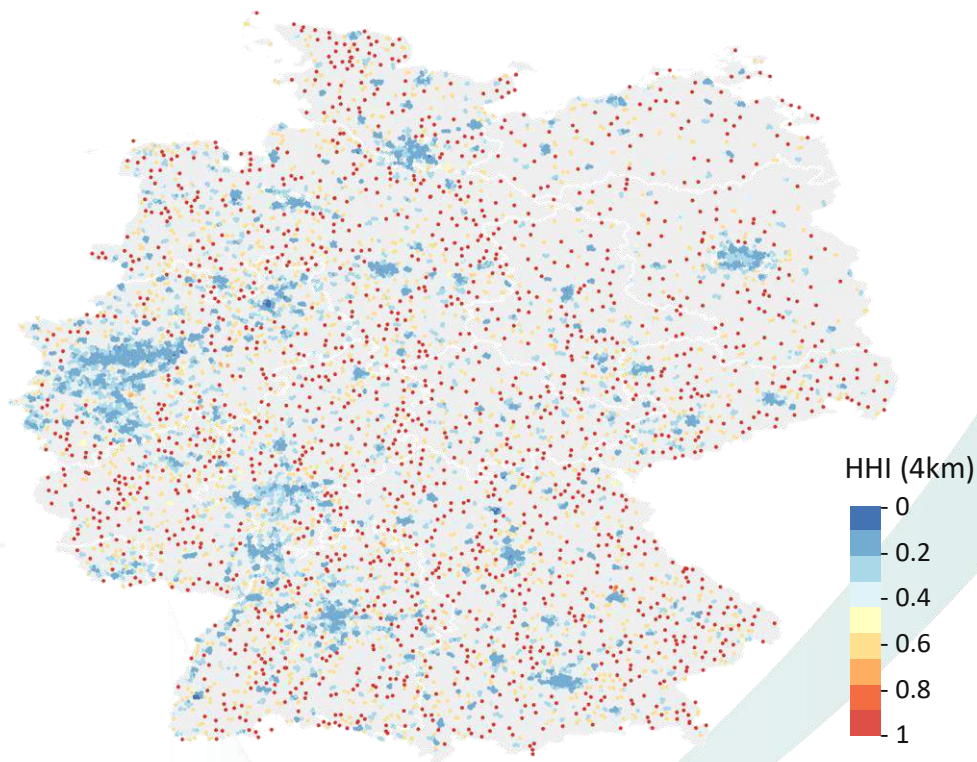
j := brand

r := radius

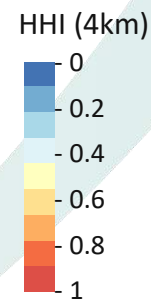
Station-level Herfindahl-Hirschman-Index (HHI) – Roadside Stations



(a) Hannover Region



(b) Germany



Summary Statistics – Market Structure

Spatial Competition (Tax Reduction Period)

		Region type		
		full sample	urban	intermediate
Distance next Comp.	1.69	1.08	1.82	2.37
	(1.93)	(1.08)	(1.99)	(2.50)
HHI	0.34	0.22	0.37	0.48
	(0.27)	(0.16)	(0.28)	(0.31)
Observations	14,257	4,791	6,451	3,015

Vertical Integration (Tax Reduction Period)

stations by segment (%)	full sample	Region type		
		urban	intermediate	rural
Oligopolistic	32.4	40.5	29.7	25.4
Other integrated	23.5	27.7	22.4	19.2
Independents	44.1	31.8	47.9	55.4
Observations	14,257	4,791	6,451	3,015

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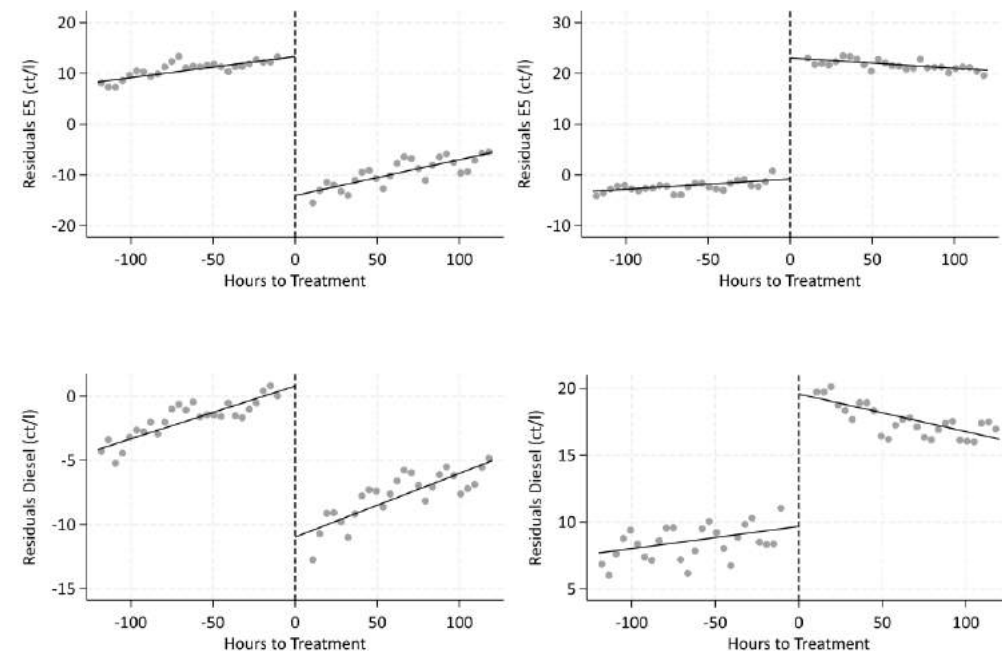
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Results: Average Pass-Through Rates of the German Fuel Discount

RDIT Plots

RDIT Results



(a) Tax Reduction

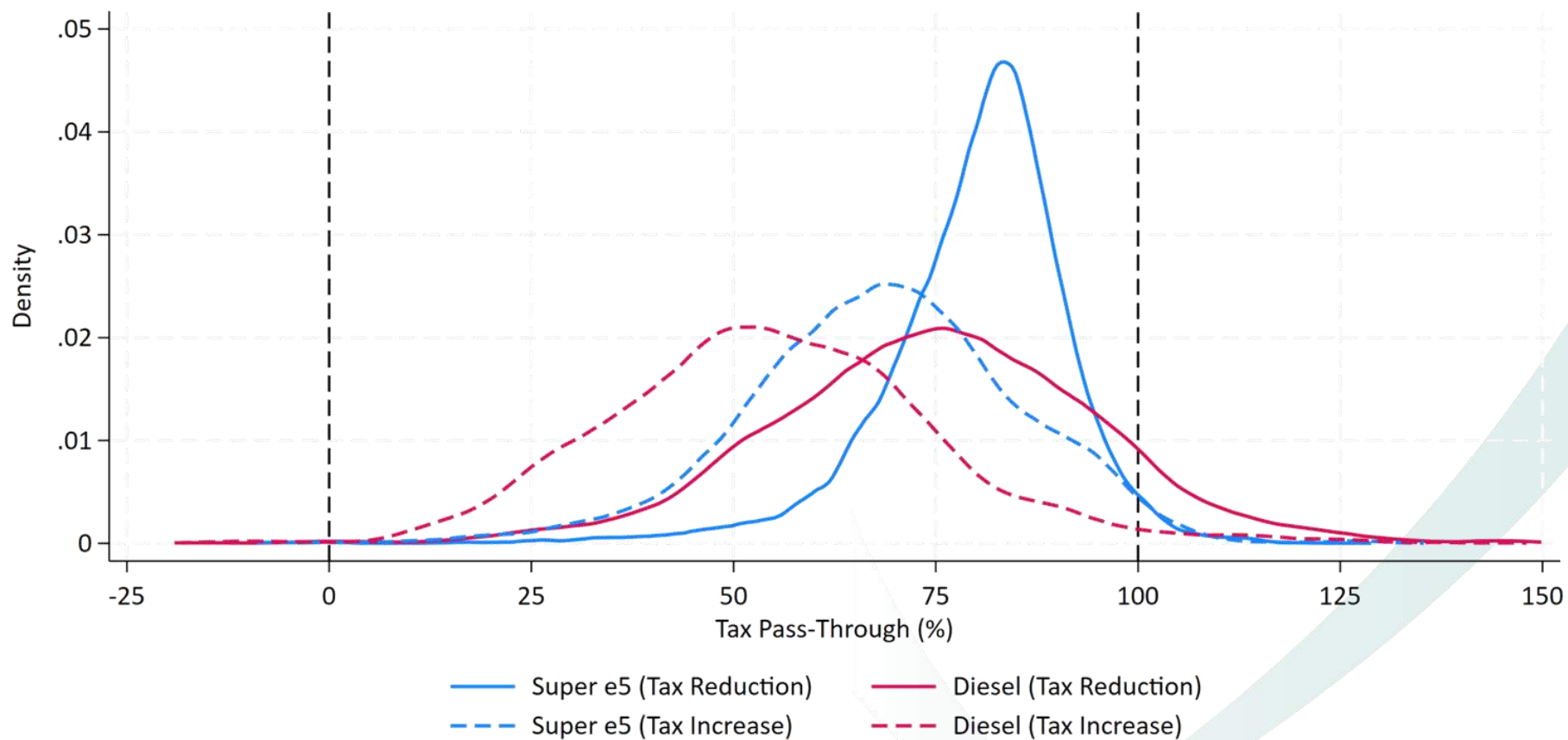
(b) Tax Increase

	Roadside Stations			
	Tax Reduction		Tax Increase	
	$\widehat{\beta}_1$	$\widehat{\rho}$	$\widehat{\beta}_1$	$\widehat{\rho}$
Super E5	-28.03***	79.7%	23.99***	68.2%
Diesel	-12.48***	74.7%	9.23***	55.2%

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

$$\widehat{\rho}^{E5} = |\widehat{\beta}_1^{E5}|/35.16; \widehat{\rho}^{Diesel} = |\widehat{\beta}_1^{Diesel}|/16.71$$

Distribution of Station-Level Pass-Through (%)



Tax Pass-Through (ct/l, %) by Market Concentration

	Tax Reduction				Tax Increase			
	Super E5		Diesel		Super E5		Diesel	
	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$
HHI Q ₁ (Ref.)	-28.73***	80.7%	-13.13***	78.6%	24.19***	68.8%	9.05***	54.2%
HHI Q ₂	0.49***	-1.4%p	0.76***	-4.5%p	0.23***	+0.7%p	0.24***	+1.4%p
HHI Q ₃	0.76***	-2.2%p	0.89***	-5.3%p	-0.30***	-0.9%p	0.11**	+0.7%p
HHI Q ₄	1.23***	-3.5%p	0.93***	-5.6%p	-0.21***	-0.6%p	0.41***	+2.5%p

* p < 0.10, ** p < 0.05, *** p < 0.01

$$\widehat{\rho}^{E5} = \widehat{\beta}^{E5} / \pm 35.16 ; \widehat{\rho}^{Diesel} = \widehat{\beta}^{Diesel} / \pm 16.71$$

Tax Pass-Through (ct/l, %) by Market Concentration

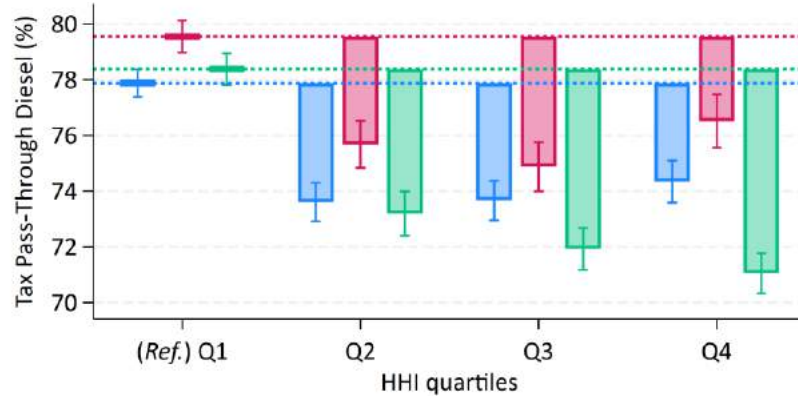
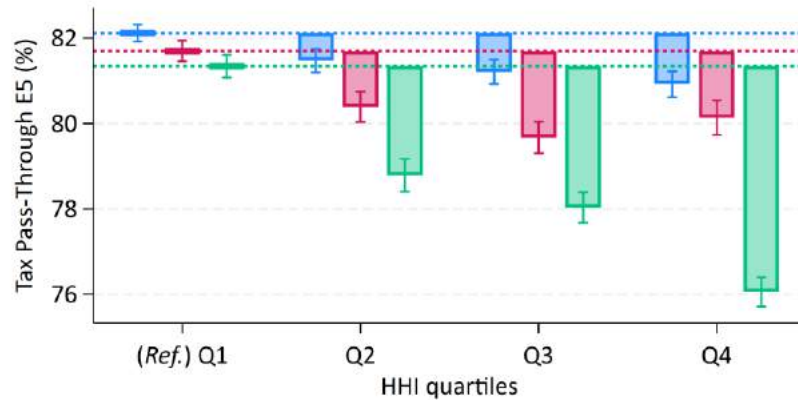
	Tax Reduction				Tax Increase			
	Super E5		Diesel		Super E5		Diesel	
	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$
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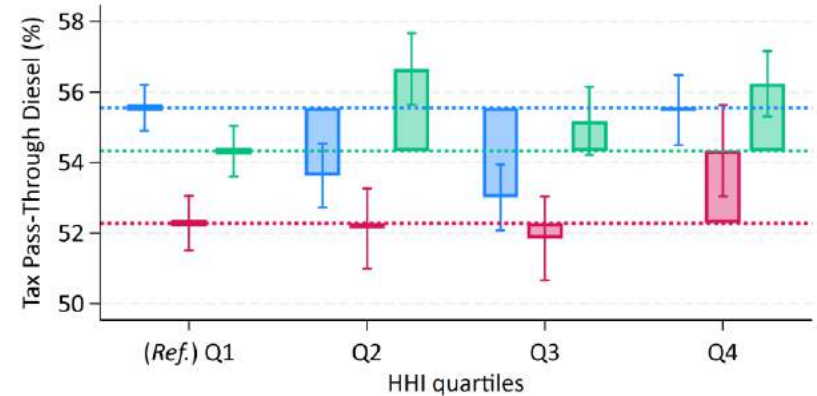
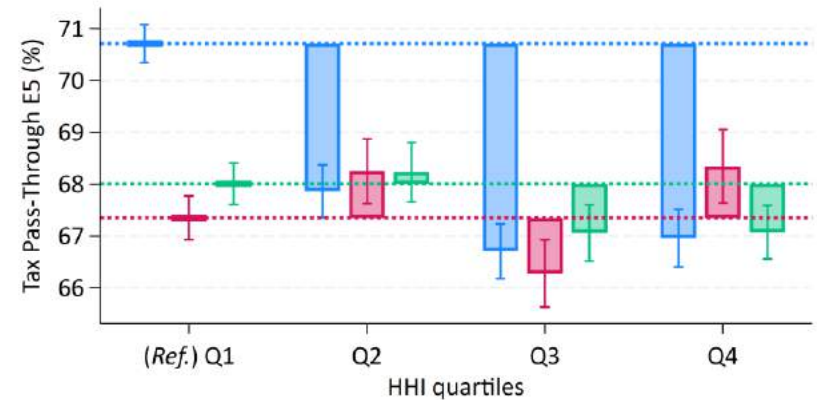
$$\widehat{\rho}^{E5} = \widehat{\beta}^{E5} / \pm 35.16 ; \widehat{\rho}^{Diesel} = \widehat{\beta}^{Diesel} / \pm 16.71$$

Pass-Through Heterogeneity – Isolating the „Competition Effect“

Tax Pass-Through (ct/l) by Vertical Integration and HHI



(a) Tax Reduction



(b) Tax Increase

Conclusions:

- **Incomplete** and **Asymmetric** Pass-Through
for the Tax Reduction (75 – 80%) and the Tax Increase (55 – 68%)
- **Market Imperfections** at the retailing level
pull the brake on Tax Pass-Through (up to 5.7%p)
- „**Competition Effect**“ on Pass-Through is ambiguous for the Tax Increase

Future Steps:

- Consistent Standard Errors by Bootstrapping both Stages
- GMM estimation for Two-Step Procedure

Thanks for your attention.



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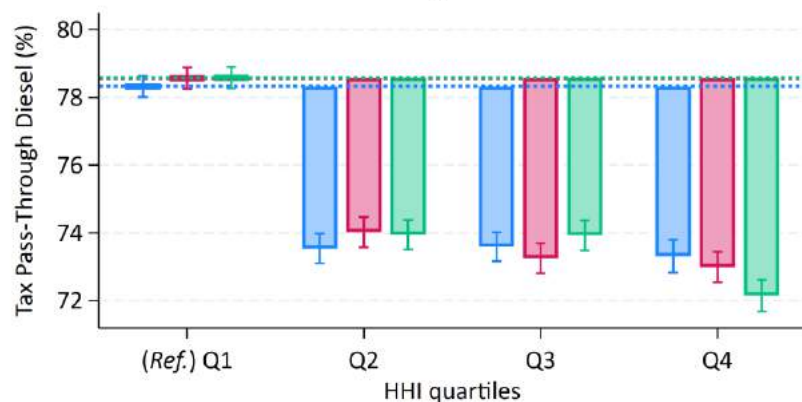
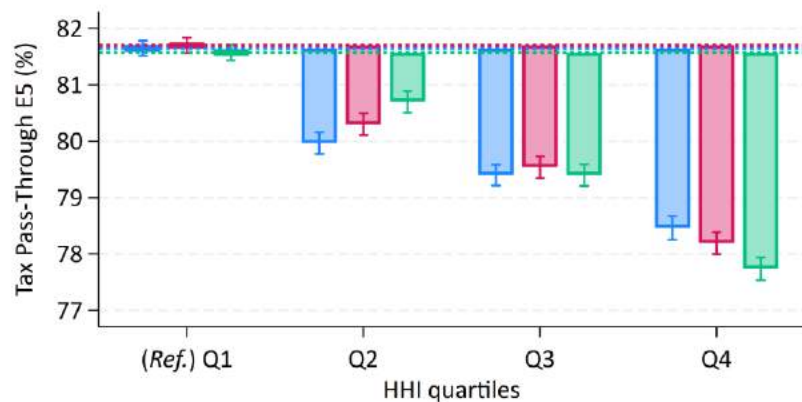
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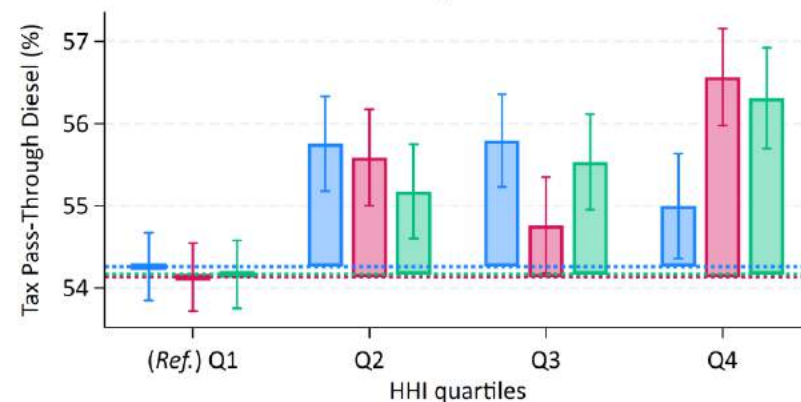
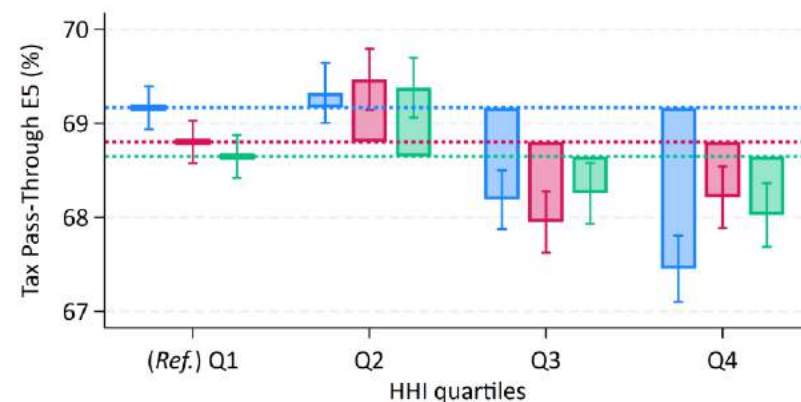
Dr. Heike Link

Backup: Sensitivity – Alternative Geographic Market Definitions

Tax Pass-Through (ct/l) by HHI and Alternative Market Demarcations



(a) Tax Reduction



(b) Tax Increase

Backup: Pass-Through Heterogeneity - Vertical Integration

Tax Pass-Through (ct/l, %) by Vertical Integration

	Tax Reduction				Tax Increase			
	Super E5		Diesel		Super E5		Diesel	
	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$	$\hat{\beta}$	$\hat{\rho}$
Oligopolistic (Ref.)	-28.64***	81.4%	-12.52***	74.9%	24.83***	70.6%	9.49***	56.8%
Other Integrated	0.31***	-0.4%p	-0.33***	-2.0%p	-1.09***	-3.1%p	-0.71***	-4.2%p
Independent	1.14***	-3.2%p	0.27***	-1.6%p	-1.10***	-3.1%p	-0.19**	-1.1%p

* p < 0.10, ** p < 0.05, *** p < 0.01

$$\widehat{\rho^{E5}} = \widehat{\beta^{E5}} / \pm 35.16 ; \widehat{\rho^{Diesel}} = \widehat{\beta^{Diesel}} / \pm 16.71$$